

We recall that the nature of a critical point (a, b) to a twice continuously differentiable function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is determined through the following result on second order optimality conditions.

Theorem: Set

$$D(a, b) := \begin{vmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{vmatrix} = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2.$$

Notice that $D(a, b)$ is the determinant of the Hessian matrix $Hf(a, b)$ defined by

$$Hf(a, b) = \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{pmatrix}.$$

- (i) If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$ or $f_{yy}(a, b) > 0$, then the critical point (a, b) is a point of local minimum for the function f .
- (ii) If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$ or $f_{yy}(a, b) < 0$, then the critical point (a, b) is a point of local maximum for the function f .
- (iii) If $D(a, b) < 0$, then the critical point (a, b) is neither a point of local minimum nor maximum for the function f . The critical point (a, b) is called in this case a **saddle** point for the function f .
- (iv) If $D(a, b) = 0$ then we have an undeterminate case for which anything can happen.

1. (a) Find all the critical points of each of the functions below:

(i) $f(x, y) = \arctan(x + y) - y^2 - x$;

(ii) $g(x, y) = \arccos(x - y) + 2x - y^2$.

(iii) $h(x, y) = 2 \sin(x - 2y) + \cos(3y)$.

- (b) Select all the critical points of the function h lying in the region

$$D := \{(x, y) \in \mathbb{R}^2 : x - 2y > 0 \text{ and } y > 0\}.$$

2. Determine the nature (points of local maximum/minimum, saddle point) of each critical point of the function

(a) $f(x, y) := 25x^3 + 12y^3 - 75x^2 - 36y$;

(b) $f(x, y) = 2x^2 - y^3 - 2xy$;

(c) $f(x, y) = x^2 - \frac{4xy}{y^2 + 1}$;

(d) $f(x, y) = e^{2x - x^2 - y^2}$.

3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \int_{-x^2}^{2y} \cos(t^2) dt \quad \forall (x, y) \in \mathbb{R}^2.$$

Find **All** the critical points of the function f .